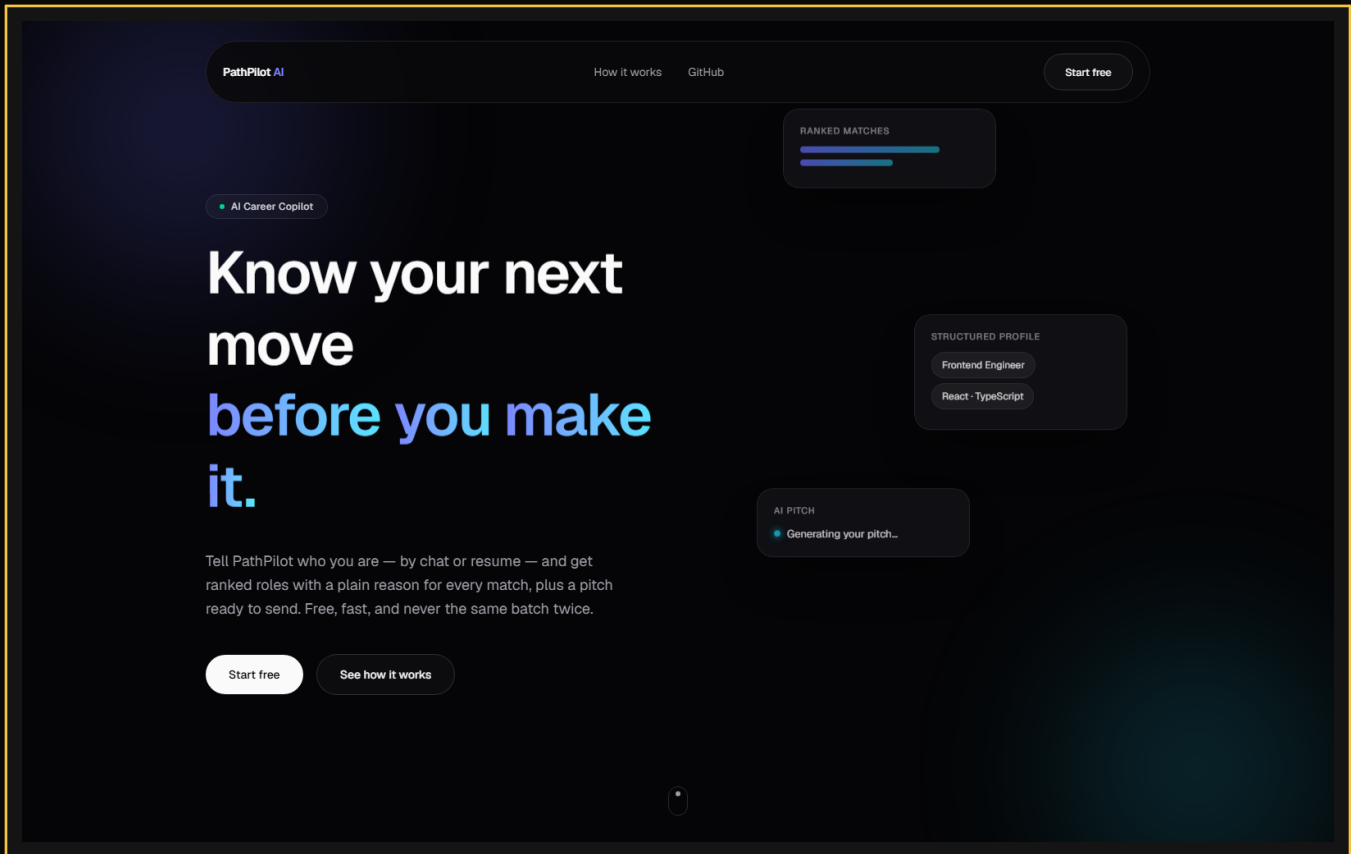


PathPilot AI

AI Career-Matching Copilot — Product Case Study



Landing page — ranked matches, structured profile, and AI pitch previewed above the fold.

OVERVIEW

Role: Sole frontend engineer — product design, UI, motion, and full-stack integration (Gemini AI, Supabase).

PathPilot AI is an AI career-matching copilot: tell it who you are by chat or resume, and it returns 8–12 ranked role matches with plain-English reasoning, plus an AI-drafted outreach pitch for the top 3. Built to demonstrate real AI integration (not a mock), motion/animation craft, and Supabase-as-BaaS competence in a single, fully working product.

PRODUCT AT A GLANCE

8–12 AI-ranked matches per batch	<60s Resume/chat to structured profile	\$0 Cost to run on the Gemini free tier
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TECH STACK

- Next.js 16 (App Router, Turbopack) + React 19
- Google Gemini 2.5 Flash via the Vercel AI SDK — generateObject for structured extraction, streamText for the live pitch
- Supabase — Postgres + Auth (magic-link, Google OAuth) + Realtime broadcast + Row Level Security
- Framer Motion, GSAP + ScrollTrigger, and Lenis for the animation layer
- React Three Fiber for an optional 3D hero (measured, shipped disabled — see engineering notes)
- Zustand for anonymous-visitor client state · deployed on Vercel · email via Resend

HOW IT WORKS

- **Tell it who you are** — a four-question chat wizard or a resume upload (PDF), visitor's choice.
- **Get a structured profile** — role interest, experience, skills, and notable projects, fully editable before anything else happens.
- **See ranked matches, with reasons** — Gemini ranks 8–12 roles from a curated pool against the specific profile, each with a one-line reason that references the visitor's actual skills.
- **Get a pitch** — for the top 3 matches, a personalized outreach pitch streams in live, ready to copy and send.
- Anonymous visitors get the full experience with zero signup, persisted to localStorage. Signing in graduates that data into Supabase, scoped per-user with row-level security.

DESIGN APPROACH

Dark UI with an indigo-to-teal gradient accent, built to feel like a live AI product rather than a static form: animated preview cards in the hero, a chat-style wizard, and a streamed pitch that types in character-by-character. Motion is used to communicate state (scanning, ranking, generating) so the wait for real AI calls reads as progress, not a stall.

ARCHITECTURE DECISIONS WORTH KNOWING ABOUT

Several build decisions were driven by real constraints rather than defaults, and are documented in the project README as a transparent record of trade-offs made along the way.

ENGINEERING NOTES

- **Gemini, not OpenAI** — the original plan called for GPT-4o. Swapped to Gemini 2.5 Flash so the whole project runs at zero cost; OpenAI requires prepaid credit with no meaningful free tier, Google AI Studio does not.
- **localStorage-first, Supabase as an upgrade path** — anonymous visitors never hit a paywall or signup wall. Signing in doesn't unlock features, it adds persistence, so a reviewer trying the site cold gets the complete experience in one sitting.
- **Custom SMTP over Supabase's default email sender** — Supabase's built-in email is rate-limited to a couple of sends per hour by design. Wired up Resend's free tier as custom SMTP so real sign-in testing doesn't get throttled.
- **The 3D hero is real, but shipped disabled** — a working React Three Fiber scene sits behind a feature flag. Measured with production Lighthouse before deciding: flag off costs nothing, flag on drops Performance from the low-90s to 49 (3.4s of blocked main thread from continuous WebGL rendering). Left disabled rather than shipped broken.
- **Live status via Supabase Realtime broadcast, no new table** — resume parsing and match ranking take several seconds. Rather than a generic spinner, the server pushes short status updates over a broadcast channel the client subscribes to per-request.
- **RLS verified directly** — Row Level Security on profiles and matches was tested by setting `request.jwt.claims` to an arbitrary UUID in the Supabase SQL editor and confirming zero visible rows, then repeating with the real signed-in user's UUID.

LINKS

- | | |
|------------------|---|
| Live Demo | https://pathpilot-ai-iota.vercel.app |
| GitHub | https://github.com/Hritwik001/pathpilot-ai |

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